

Clinical Trials IV Assignment 1

February 2026

1 Introduction

This report outlines a randomised controlled trial for a treatment for Ventrothermal Drift. The main symptom of this disease is overestimation of torso temperature, while it also causes patients to feel lethargic and generally unwell. The population will consist of participants between the ages of 50 and 65, so any conclusions will be limited to this age range, as it is not representative of the general population. The primary outcome variable will be the temperature differential as this is the primary symptom of the condition. The protocol detailing allocation and analysis should be defined ahead of recruiting participants. Particularly as the primary outcome variable is subjective, it is essential to keep clinicians and participants blind to which arm they are on, otherwise assessment bias may change the results and invalidate the trial. To ensure the trial is ethical, clinicians should monitor participants over the course of the trial to ensure that they are not harmed or experiencing adverse side effects as a result of the treatment.

We test the hypotheses:

$$\begin{aligned}\mathcal{H}_0 : \tau &= 0, \\ \mathcal{H}_1 : \tau &\neq 0,\end{aligned}\tag{1}$$

where τ represents the treatment effect.

2 Sample Size

The power of a two-sided t-test is given by:

$$\Psi(\tau) = 1 - \beta = \left[1 - \Phi\left(z_{\alpha/2} - \frac{\tau_M}{\sigma_\lambda}\right)\right].\tag{2}$$

We assume that we have samples of equal size (N) in each arm, as this minimises the standard error and therefore maximises the statistical power of the test. Under this assumption, we have $\lambda = \sqrt{2/N}$. As we would like to be able to detect an effect size of at least 2°C, with a power of 90% and at a significance level of 0.05, we have:

$$\begin{aligned}\tau_M &= 2, \\ z_{\alpha/2} &= z_{0.975} = 1.96, \\ \Psi(\tau) &= 0.9.\end{aligned}$$

We also have that the temperature differential has a standard deviation of approximately 4.5°C for both treatment and control groups. Substituting in eq. (2) and rearranging gives:

$$\begin{aligned}N &= \frac{2\sigma^2 (z_\beta + z_{\alpha/2})^2}{\tau_M^2} \\ &= 106.29.\end{aligned}$$

Therefore, the minimum sample size required per arm of the trial would be 107. However, we also know that the correlation (ρ) between baseline and endline temperature differential measurements will be around 0.7. As a result, we should use ANCOVA analysis to account for this correlation. We know that the variance in ANCOVA analysis is lower than that of a standard t-test:

$$\text{Var}(\hat{\tau}_{\text{ANCOVA}}) = \frac{2\sigma^2}{N}(1 - \rho^2) < \frac{2\sigma^2}{N} = \text{Var}(\hat{\tau}_{\text{t-test}}).\tag{3}$$

We can therefore use the fact that “ANCOVA with $(1 - \rho^2)n$ subjects has the same power as t-test with n subjects” (Borm, Fransen, and Lemmens 2007), allowing us to use fewer participants. This is not only more ethical, as fewer patients receive a placebo as opposed to a real treatment, but also cheaper, since recruiting fewer patients will be faster and require less resources. Therefore the adjusted sample size becomes:

$$\begin{aligned}N &= \lceil 106.29(1 - 0.7^2) \rceil \\ &= \lceil 54.21 \rceil \\ &= 55.\end{aligned}$$

To ensure we maintain the power of the trial, we should also account for potential participant attrition. This reduces the likelihood that we will have to rerun the trial due to insufficient power, which is expensive and hinders the release of a potentially beneficial treatment. Based on an anticipated retention rate of 88% - the median retention rate observed in a large-scale review of RCTs (Table 4, Jacques et al. 2022) - we calculate the new sample size per arm as:

$$N = \left\lceil \frac{54.21}{0.88} \right\rceil = 62.$$

Now that we have computed our a sample size per arm, we must decide how to allocate participants to each arm.

3 Allocation

3.1 Stratification

We measure five baseline covariates, and want to ensure that the trial arms are roughly balanced with respect to each covariate to prevent confounding. Simultaneously, we want to avoid creating an unworkable number of strata, each containing very few patients.

Since patients suffering from Ventrothermal Drift tend to overestimate their torso temperature, and our primary outcome variable is the temperature differential, it is important to consider the effects of each covariate on temperature perception. Women can be “more sensitive to temperature fluctuations”, and are “more likely to be hot or cold than men” (Schweiker et al. 2018), indicating that stratifying by sex may be important. Additionally, one would expect that on average the weight and RMR would be higher for men than women. In this way, sex may act as a proxy for weight and RMR, implicitly balancing them across the trial arms. Although we do not anticipate a biological correlation between sex and age, and age may be a factor affecting temperature perception (Schweiker et al. 2018), the population consists of a narrow age range (50 – 65 years) which should account for any major differences. Therefore we stratify based on sex alone. Stratifying further may be redundant and would involve creating buckets from continuous data which would reduce precision. It also increases the risk of having many strata, with some containing very few participants. Since we will use an ANCOVA model, the effects of other covariates can be adjusted for as part of the analysis.

3.2 Random Permuted Blocks

We use randomly permuted blocks to allocate participants to trial arms since this ensures fairly balanced numbers in each arm, which is not guaranteed by methods such as simple random sampling, for example. Random permuted blocks is a simple, reproducible algorithm which works well for our single binary stratification of sex. We use random block lengths of 2, 4, 6 and 8 to mitigate the possibility of allocation being predicted if, for any reason, the trial blindness is compromised, thus maintaining the integrity of the trial. A potential disadvantage of using different block lengths is that there is increased chance of imbalance in the sample sizes of the trial arms. The maximum theoretical imbalance would be a sample size difference of 8 between the control and trial arms, which would occur if the final block for both strata was of length 8 and of the form “CCCCTTTT” or “TTTTCCCC”. This is very unlikely:

$$\begin{aligned} \mathbb{P}(\text{block length } 8) &= \frac{1}{4}, \\ \mathbb{P}(\text{CCCCTTTT}) + \mathbb{P}(\text{TTTTCCCC}) &= 2 \cdot \frac{1}{2} \cdot \frac{3}{7} \cdot \frac{2}{6} \cdot \frac{1}{5} = \frac{1}{35}, \\ \mathbb{P}(\text{Imbalance of 4 in one block}) &= \frac{1}{4} \cdot \frac{1}{35} = \frac{1}{140}, \\ \mathbb{P}(\text{Total imbalance of 8}) &= \frac{1}{140^2} = \frac{1}{19600}. \end{aligned}$$

Moreover, this imbalance is unlikely to affect the power significantly. In a trial of total size 124, a difference of 8 would lead to 58 in one arm and 66 in the other. This is unlikely to have a significant effect

on the power of our test since:

$$\lambda(58, 66) = \sqrt{\frac{1}{58} + \frac{1}{66}} \approx 0.1800,$$

$$\lambda(62, 62) = \sqrt{\frac{2}{66}} \approx 0.1796.$$

These values only differ by 0.0004, and so from eq. (2), we see that the difference in power will be very small. Therefore, the benefits of using randomly permuted blocks outweigh the risks.

3.3 Allocation Results

Though the allocation process was strictly predetermined to allow sequential recruitment into the trial and prevent selection bias, we can review the baseline measurements after allocation is complete to confirm that structural balance was achieved. In Table 2, we see that the arms are closely balanced in size (imbalance of 2) and that we have successfully stratified by gender with a marginal imbalance of 2. Moreover, the results show that the weight, RMR and baseline temperature differential exhibit no meaningful imbalances, and align closely with the population means from Table 1.

Characteristic		N=124
Sex	F	77 (62%)
	M	47 (38%)
Age (Years)		57.4 (53.5, 61.3)
Weight (kg)		73 (66, 81)
RMR (kcal/day)		1520 (1338, 1710)
Baseline Temperature Differential (°C)		9.3 (7.0, 12.5)

Table 1: Population characteristics, presented as *count (proportion)* for factor variable sex, and *median (Q1, Q3)* for all continuous variables.

Characteristic		C	T
N		63	61
Sex	F	39 (62%)	38 (62%)
	M	24 (38%)	23 (38%)
Age (Years)		56.7 (53.5, 61.2)	58.0 (53.5, 61.3)
Weight (kg)		73 (66, 82)	74 (67, 79)
RMR (kcal/day)		1,525 (1,334, 1,711)	1,517 (1,352, 1,706)
Baseline Temperature Differential (°C)		10.0 (7.1, 13.0)	9.0 (6.9, 11.4)

Table 2: Population characteristics by trial arm, presented as *count (proportion)* for factor variable sex, and *median (Q1, Q3)* for all continuous variables. Column C represents the control arm of the trial, and column T represents the treatment arm.

We confirm this further with Figure 1, indicating that stratifying by sex implicitly encourages balance in the split of weight (Figure 1b) and RMR (Figure 1c) covariates across the trial arms, as expected. We note that, as expected, stratifying by sex has no balancing effect on age, and so age is allocated randomly across the trial arms (Figure 1d). Overall, the combination of stratification by sex and randomly permuted blocks led to relatively balanced trial arms, from which remaining imbalances can be handled in the analysis with the ANCOVA model.

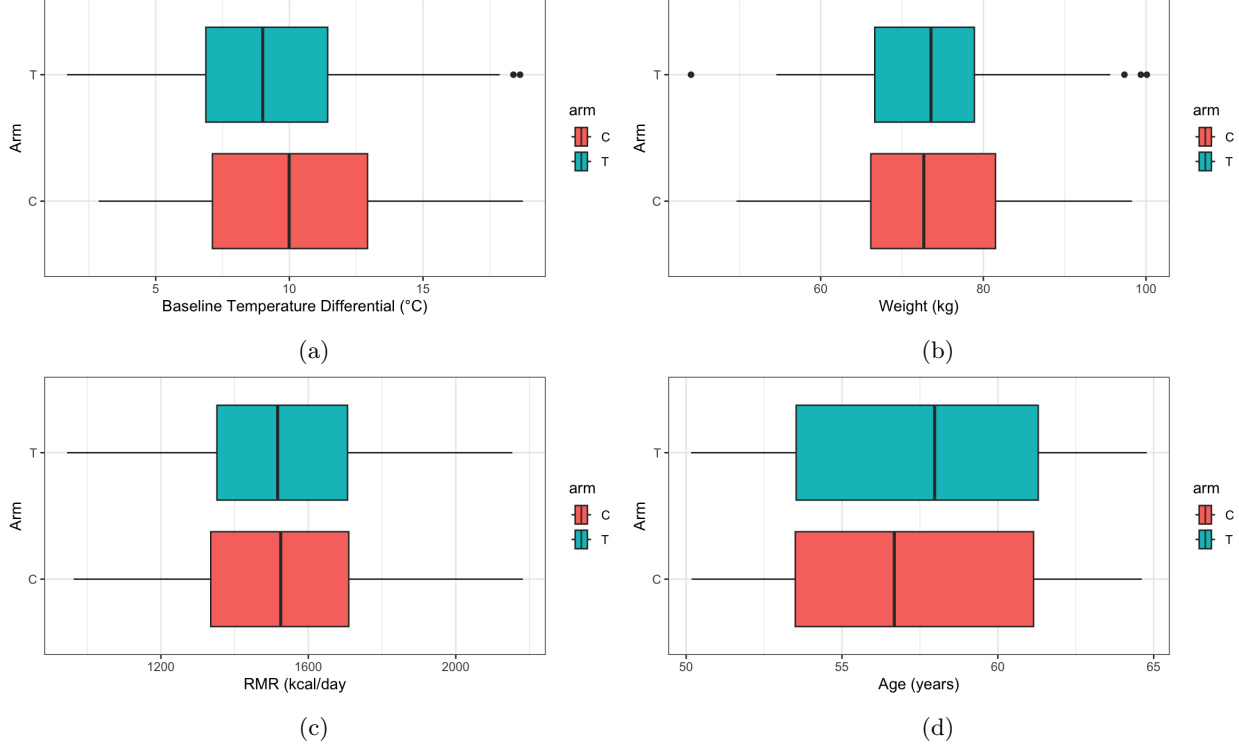


Figure 1: Box plots by trial arm (control - C, red, treatment-T, blue) of the continuous baseline covariates: baseline temperature differential (a), weight (b), resting metabolic rate (c) and age (d).

4 Analysis

As shown in Figure 1a, the mean and variance of the baseline temperature differential are not identical across trial arms; by random chance we have a mean difference of 1.0°C (Table 2). We also observe that the spread of the control group is greater than that of the treatment group. Though not necessarily indicative of an imbalance, it is important to account for this difference since we have strong correlation between baseline and outcome measurements, with $\rho \approx 0.7$. As shown in eq. (3), using an ANCOVA model will reduce the variance, performing better than a t-test since we have strong correlation ($\rho > 0.5$).

If we had some attrition and therefore no endline temperature differential measurement for some participants, these participants would be removed from the final model, which could impact the power if the attrition rate is higher than the 12% accounted for in Section 2. Though there was no sample attrition here, in practice we should be aware of this as it can introduce bias if multiple people with similar characteristics drop out. We should investigate to ensure attrition is random and not linked to a specific characteristic or arm of the trial, so that the treatment effect is not skewed.

4.1 Methodology

The analysis is defined as an ANCOVA model adjusting for all covariates (treatment, sex, RMR, weight, age). However, biologically, we believe that weight and RMR are likely to be correlated (Argyraopoulou et al. 2023). We can check for multicollinearity by calculating the Variance Inflation Factor (VIF):

$$VIF(\hat{\beta}_j) = \frac{1}{1 - \mathbf{R}_j^2}, \quad (4)$$

where $\mathbf{R}_j$ is the squared multiple correlation for the regression of x_j on all other covariates.

Without seeing the data, we can account for possible multicollinearity by deciding that if one or more covariates in the model has a variance inflation factor greater than 3.0, the covariate with largest VIF is

removed from the ANCOVA linear model and the VIF is recalculated. This should be repeated until all covariates have $VIF < 3.0$. This criteria is decided before the trial in order to prevent selection bias, whilst still adjusting for multicollinearity. The exception is that variables used for stratification should always be included in the ANCOVA model, and so sex should not be removed, even if its VIF is high. This accounts for the ‘non-randomness’ introduced by the constraint of balancing arms by sex.

We do not include interaction terms of covariates in the linear model, since we do not believe a-priori that the drug behaves differently for different weights, ages or sexes. If there was a biological reason that, for example, the drug would work better on women, then we would introduce an interaction term between sex and treatment arm.

4.2 ANCOVA Results

We firstly compute the VIF for each of the baseline variables. We see in Table 3 that both RMR and weight have VIF values greater than the threshold of 3.0. This indicates multicollinearity and so we remove resting metabolic rate (largest VIF) as a covariate from our model. After recalculating the VIF of the new model, all variables have VIF less than the threshold, and so we include them in our model.

Variable	VIF
Baseline Temperature Differential	1.365
Sex	1.292
Age	1.562
Weight	3.354
Resting Metabolic Rate	3.868

Table 3: Variance Inflation Factor for each variable when including all baseline variables in the model.

Variable	VIF
Baseline Temperature Differential	1.281
Sex	1.250
Age	1.041
Weight	1.480

Table 4: Variance Inflation Factor for each variable after removing resting metabolic rate from the model.

This results in a linear model of the form:

$$\delta_{endline,i} = \beta_0 + \tau G_i + \beta_1 \delta_{baseline,i} + \beta_2 a_i + \beta_3 s_i + \beta_4 w_i + \epsilon_i, \quad (5)$$

where $\epsilon_i \stackrel{i.i.d}{\sim} \mathcal{N}(0, \sigma_\epsilon^2)$, δ represents the temperature differential, G_i is an indicator variable with $G_i = 1$ for participants in arm T and $G_i = 0$ for those in arm C. δ represents the temperature differential and a , s , and w represent the covariates age, sex and weight respectively, with s a factor (0 for female and 1 for male). $\beta_i \in \mathbb{R}$ are the coefficients of the linear model which we are estimating and $i = 1, \dots, 124$ represent the participants of the trial.

Using this model, we see the results for the coefficients as in Table 5. Based on these results, we have sufficient evidence to reject the null hypothesis in eq. (1), since our p-value (3.40×10^{-7}) is much less than our significance level ($\alpha = 0.05$). This gives a 95% confidence interval for the treatment effect τ as:

$$-4.481 \pm t_{0.975;118} \times 0.8287 = [-6.12, -2.84]. \quad (6)$$

Since this is entirely negative and does not contain zero, we can be confident that the treatment decreases the temperature differential.

We have a low p-value for the baseline temperature differential, which we would expect since this indicates it is a good predictor for the endline temperature differential. Including it in the ANCOVA analysis reduces the residual variance, therefore increasing the precision of the estimate of the treatment effect. We also see that age is significant ($p=0.00402$), with its positive coefficient indicating that as age increases, the disparity in temperature perception also increases. Weight and sex were not found to be significant covariates. The model had an adjusted R^2 value of 0.528, indicating that 52.8% of the variance in the data is explained by the covariates, which is fairly strong given the nature of the trial, as there are many other uncontrolled variables, including sleep, diet and other illness. The standard errors are fairly small (< 1 for all covariates) indicating precise estimates. For a deeper analysis we look at the residuals.

We see that the residuals are roughly centred around zero for both arms (Figure 2), indicating that the model is not biased towards either group. The residual variance is greater for the control group, suggesting

	Estimate	Std. Error	p-value
Intercept	-15.71	6.735	0.02139
Baseline Temperature Differential	0.9757	0.1153	8.24×10^{-14}
Sex	-0.5689	0.9444	0.5481
Age	0.2831	0.09649	0.00402
Arm	-4.481	0.8287	3.40×10^{-7}
Weight	0.01114	0.04563	0.8112

Table 5: Result of applying linear model (5) to the data.

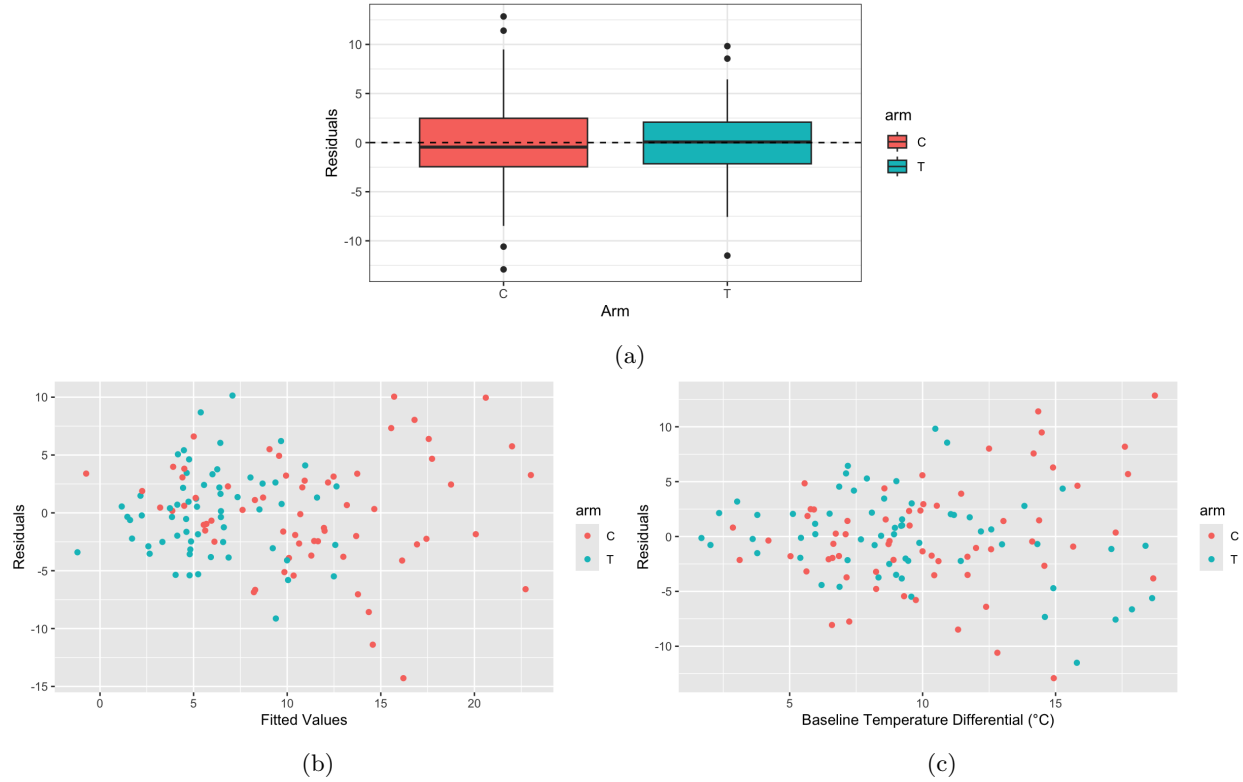


Figure 2: Plots for residuals of model (5) applied to the data, with control arm (red) and treatment arm (blue). (a) Box plot of residuals by trial arm. Plot of residuals against fitted values (b) and baseline temperature differential (c).

that the drug may have a stabilising effect. In the control group, the recovery can depend on various individual and lifestyle factors, hence the spread of outcomes, whereas in the treatment group the outcomes are more predictable. When comparing the residuals with the baseline temperature differential, the points form a clear cone pattern, with increasing residual variance as baseline temperature differential increases. This may be partially influenced by a higher density of treatment participants at lower baseline values, and the higher variance in the initial control group. However, it may also suggest that the treatment effect could depend on the baseline temperature differential. To account for this, we consider an interaction term in our model between the trial arm and the baseline temperature differential. Our new model is of the form:

$$\delta_{endline,i} = \beta_0 + \tau G_i + \beta_1 \delta_{baseline,i} + \beta_2 a_i + \beta_3 s_i + \beta_4 w_i + \beta_5 \delta_{baseline,i} G_i + \epsilon_i, \quad (7)$$

This allows for the control and treatment linear models to have different gradients. We see in Table 6 that the interaction term is significant at the 5% level, with p-value 0.00108, which indicates that the treatment effect depends on the baseline temperature differential. This gives a 95% confidence interval for the interaction term as:

$$-0.6675 \pm t_{0.975;117} \times 0.1991 = [-1.06, -0.273].$$

Again, this is entirely negative and so we can be confident that the ‘*treatment:baseline*’ interaction term decreases the temperature differential. It suggests that for those with more severe baseline temperature differential, the drug will be more effective. This model has adjusted R^2 of 0.5656, which suggests that it is an improvement from the previous model, as more of the variability in the data is explained by the covariates.

We see in Figure 3 that the residual variance still increases with baseline temperature differential, with a clear shift in variance at around $10^\circ C$. We note from Figure 3a that the mean residuals under the new model are more balanced between arms. Under the new model, the treatment variance appears smaller, but not the control group. This idea is amplified by Figure 3b, where the treatment arm are clustered at lower fitted values, more tightly packed and the control arm fitted values are scattered randomly with high variance. This suggests that the assumption of σ_ϵ being constant is not satisfied in practice, and that there is some heteroscedasticity. This can be interpreted as the biological variation in the control group, as untreated participants’ recoveries could vary greatly based on external factors, possibly suggesting the presence of unmeasured confounding variables which impact the natural recovery from the disease. Furthermore, the heightened residual variance may also be a result of the subjective nature of the results. Since the data is based on perception, patients are likely to have varying degrees of accuracy in estimating their torso temperature, increasing variance in the data, as opposed to an objective measurement carried out by the same clinician.

	Estimate	Std. Error	p-value
Intercept	-17.12	6.474	0.00930
Baseline Temperature Differential	1.340	0.1551	3.36×10^{-14}
Arm	2.136	2.128	0.3176
Sex	-0.3441	0.9085	0.7056
Age	0.2694	0.09265	0.00436
Weight	-0.01120	0.04513	0.8044
BaselineTempDiff*Arm	-0.6675	0.1991	0.00108

Table 6: Result of applying linear model (5) to the data.

4.3 Secondary Analysis

We can analyse the effect of the treatment on wellness score using a logistic model. Those suffering from Vetrothermal Drift feel generally unwell, and so by including a wellness score we can gain insight as to whether the drug alters only the torso temperature perception or also helps patients to feel better overall. This was evaluated with a binary logistic regression model, with results given by Table 7. The treatment

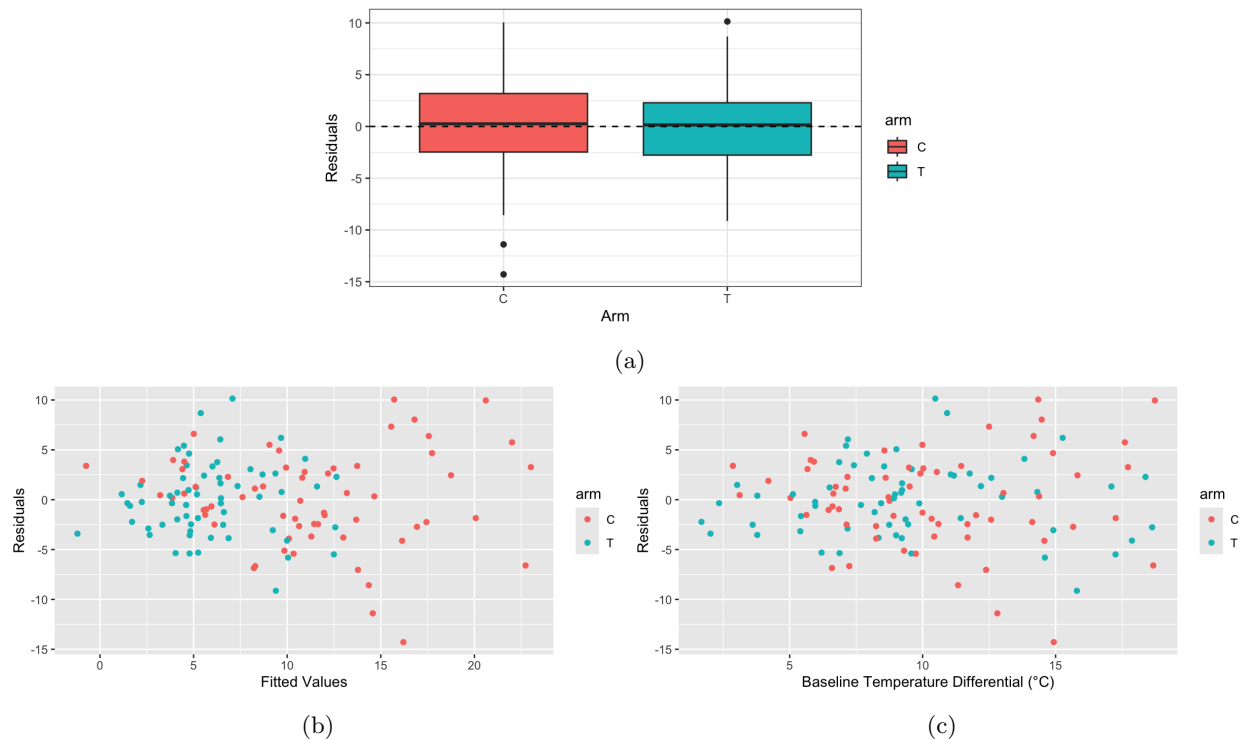


Figure 3: Plots for residuals of model (7) applied to the data, with control arm (red) and treatment arm (blue). (a) Box plot of residuals by trial arm. Plot of residuals against (b) fitted values and (c) baseline temperature differential.

is found to have a significant effect on wellness score, with a p-value of 0.000525. The coefficient for trial arm is 1.4, indicating that participants taking the drug were approximately 4 times more likely to feel an improvement in their overall wellness (odds ratio = 4.055). Interestingly, weight is also significant here, though it did not impact the temperature differential. Since it has a negative coefficient, the more a patient weighed the less likely they were to have an improvement in wellness. One reason for this may be that the treatment dosage should be adjusted based on weight, and so participants who weighed more felt less improvement. This is dependent on the nature of the treatment and likely requires further investigation.

These results should be used to complement the analysis of the primary outcome variable, since the trial's power and hypothesis were constructed based on the temperature differential alone. Therefore conclusions regarding the treatment should be made based on the primary analysis, with the secondary analysis added only to provide qualitative context about the patients' wellbeing. The significant increase in odds of improved wellness indicates a holistic recovery, rather than isolating the symptom of temperature perception.

	Estimate	Std. Error	p-value
Intercept	6.434	3.345	0.05444
Arm	1.400	0.4038	0.000525
Baseline Temp Diff	-0.007894	0.05515	0.886178
Age	-0.06168	0.04713	0.1907
Sex	-0.5559	0.4635	0.2304
Weight	-0.04606	0.02285	0.04382

Table 7: Results of the logistic model on wellness score.

5 Conclusion

In conclusion, we see that the treatment has a statistically significant effect on the temperature differential (Table 5), and so is an effective treatment for Ventrothermal Drift. However, there is evidence that the effectiveness of the treatment is dependent on the initial temperature differential (Table 6). In other words, those with a greater initial overestimate of their torso temperature see a greater improvement as a result of the treatment. In both models, age appeared to significantly effect the outcome. This variance was accounted for by the model, however the population here is from a very specific age group (50-65 years). Further trials would be needed to assess the impact of age on the outcome for other age groups. Weight and sex were not significant in the temperature differential models, however the secondary analysis indicated that participants with a higher weight had a smaller improvement in wellness. In general, the treatment led to improved wellness scores, however this may not be sufficiently powered since the trial was designed for the primary outcome variable of temperature differential. Despite this, the complementary insight from the wellness score contextualises the improvement and suggests that the treatment aided the additional symptoms of the disease. The trial could be repeated to test for heteroscedasticity, as the residual variance seemed to increase with baseline temperature differential in the ANCOVA analysis. Another limitation is the self-assessment of torso temperature, which can introduce more variance into the model. If there is a more objective measure associated with any of the symptoms, this may produce more consistent results across patients.

References

- Argyropoulou, Georgia et al. (2023). “Revisiting Resting Metabolic Rate: What is the Relation to Weight Fluctuations?” In: *Current Obesity Reports* 12.4, pp. 502–513.
- Borm, George F., Jaap Fransen, and Wim A. J. G. Lemmens (2007). “A simple sample size formula for analysis of covariance in randomized clinical trials”. In: *Journal of Clinical Epidemiology* 60.12, pp. 1234–1238. DOI: 10.1016/j.jclinepi.2007.02.006. URL: <https://doi.org/10.1016/j.jclinepi.2007.02.006>.
- Jacques, Richard M et al. (2022). “Recruitment, consent and retention of participants in randomised controlled trials: a review of trials published in the National Institute for Health Research (NIHR) Journals Library (1997–2020)”. In: *BMJ Open* 12.2. ISSN: 2044-6055. DOI: 10.1136/bmjopen-2021-059230. eprint: <https://bmjopen.bmj.com/content/12/2/e059230.full.pdf>. URL: <https://bmjopen.bmj.com/content/12/2/e059230>.
- Schweiker, M. et al. (2018). “Drivers of diversity in human thermal perception - A review for holistic comfort models.” In: *Temperature (Austin, Tex.)* 5.4, pp. 308–342. URL: <https://doi.org/10.1080/23328940.2018.1534490>.